Pipeline Architecture.

Title Page

PW SKILLS

PIPELINE ARCHITECTURE

Project:
Cryptocurrency Historical Prices Prediction using Machine Learning

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Introduction

The Pipeline Architecture describes the complete workflow of the Cryptocurrency Historical Prices Prediction project. It illustrates how raw historical cryptocurrency data is transformed into meaningful predictions through preprocessing, feature engineering, machine learning, and model evaluation.

Pipeline Flow

Dataset

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Data Loading

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Data Cleaning

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Exploratory Data Analysis (EDA)

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Feature Engineering

↓

Feature Scaling

↓

Train-Test Split

↓

Random Forest Regressor Model

↓

Model Evaluation

↓

Prediction

↓

Save Model (.pkl)

↓

Prediction Results (.csv)

Pipeline Components

Data Loading
The historical cryptocurrency dataset is imported using the Pandas library.

Data Cleaning
Missing values and invalid records are checked and handled before training.

Feature Engineering
New features such as Price Change, MA7, and Volatility are created.

Feature Scaling
StandardScaler is used to normalize numerical features.

Model Training
Random Forest Regressor is trained using the processed dataset.

Model Evaluation
Performance is measured using MAE, RMSE, and R^2 Score.

Prediction
The trained model predicts cryptocurrency volatility.

Model Saving
The trained model is saved using Joblib for future use.

Advantages of Pipeline

• Organized workflow
• Easy to maintain
• Faster model training
• Better preprocessing
• Reusable machine learning pipeline
• Deployment ready architecture

Conclusion

The pipeline architecture provides a structured and efficient workflow for cryptocurrency volatility prediction. Each stage contributes to improving data quality, model accuracy, and overall project reliability.